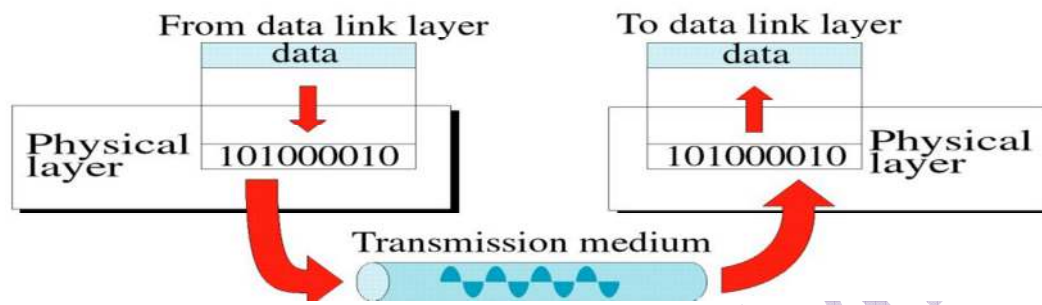


Chapter 2

Physical Layer

Functions of physical layer:-



Physical layer is the lowest layer of the OSI reference model. All the physical connectivity of this layer is takes place.

The following are the functions of physical layer of the OSI model

Physical characteristics of media and its types:- It defines the physical characteristics and functions of the physical devices so that transmission can occur. It may be wired or warless.

Representation of bits:- The physical layer data involves a stream of bits (sequence of 0's and 1's) without any interpretation. To be transmitted bits must be encoded into the signals - electrical or optical.

Data rate:- This layer defines the rate of transmission which is the number of bits per second.

Synchronization of bits:- It is necessary to have synchronization between sender and receiver at the bit level that is the clocks of the sender and the receiver must be synchronized.

Line configuration:- There are two types of line configuration

1. Point to Point configuration:- In Point-to-Point configuration, there is link between two devices.
2. Multi-Point configuration:- In Multi-Point configuration, there is a line through which multiple devices are connected.

Transmission Mode:-There are three types of transmission of mode.

1. Simplex:- Communication is unidirectional that is data can be sent only in one direction. Example- Input from keyboards, monitors, TV broadcasting, Radio broadcasting, etc.
2. Half duplex:- both devices can send and receive the data but only one at a time not simultaneously. Example- Walkie-Talkie.

3. Full Duplex:- In this mode, both devices can send and receive the data simultaneously.
Example- Telephone System, Chatting applications, etc.

HUB

A hub is one of the basic components in a computer network. It works in the physical layer of OSI model. A hub does not have decision making capability and, therefore, is considered as a non-intelligent device. Its main function is to connect multiple network hosts. Basically, a hub takes the data in one of its ports and copies that data packet to send it to all of the other ports connected to the network. That means a hub sends all the data it receives to the ports. As a hub does not have decision making the power it is not capable to find out the correct path for data packets. This, in turn, leads to inefficiencies in the network.

There are mainly two types of hubs;

Active Hub — An active hub is considered smarter than its counterpart — passive hub. This hub has its own power supply and it not only provides a path for data packets but also clean, strengthen, and concentrate the data signals before sending them to the port they belong. If we want to increase the distance between nodes then these are the perfect ones. An active hub is also known as a ‘repeater’.

Passive Hub — A passive hub does not modify, strengthen, or boost the incoming data signal. These hubs work as a meeting point for all the wires coming from nodes and the power supply from an active hub. We cannot extend the distance between nodes using the passive hubs.

SWITCHES

A switch is basically a data link layer device. As a switch has decision making capability it is considered to be one of the intelligent devices available on a computer network. A switch has the knowledge of MAC addresses of ports connected in the network and, therefore, it enhances the networking performance constantly. A switch works in full-duplex mode and also provides security as it sends the data packets only to specific and desired locations. Additional security features like firewalls can be added to a switch.

BRIDGE

A bridge is a computer networking device which operates at data link layer and connects other bridge networks using the same protocol. A bridge can connect two LAN connections working on the same protocol. Basically, it is just like a bridge on a river which connects one bank to the other. There are two types of bridges;

REPEATER

A repeater can also be called as a booster. It works at the physical layer in a computer networking system and its main purpose is to regenerate the signal over the same network before it gets weak and diminishes. This helps to extend the signal's travel length. A repeater, however, is not a signal amplifier. When the signal gets weak it simply copies it bit by bit and sends it to the route. A repeater is similar to a Hub but with added functionality. Both repeaters and hubs, face one mutual problem, i.e. only one transmission at a particular time. In case of data transmission through several devices, the data collides and gets lost.

MODEM

The word is the combination of Modulation and the Demodulation operations. A modem is a device used to connect any computer to an internet connection. A modem converts digital signals into analog signals so that it can travel through telephone lines. There are two types of modems available; internal and external modems. Generally, external modems are used mostly. An internal modem is just like a memory card called add-in expansion cards and PCMCIA cards used for laptops.

ROUTER

Routers are layer-3 devices in the OSI model. A router is used to route the traffic from one network to another. A router works to direct the traffic in a different direction. A router is used for creating complex networks by complex traffic routing. A router connects LANs and WANs together and divides broadcast domains of hosts connected with it.

A router is an intelligent device as it reads the destination address from the data packet and routes it to the location. A router is able to send the data packet to its desired location by searching in its routing table. Therefore, a routing table must be updated and complete to avoid data collision and loss. A router essentially receives information through two ways:

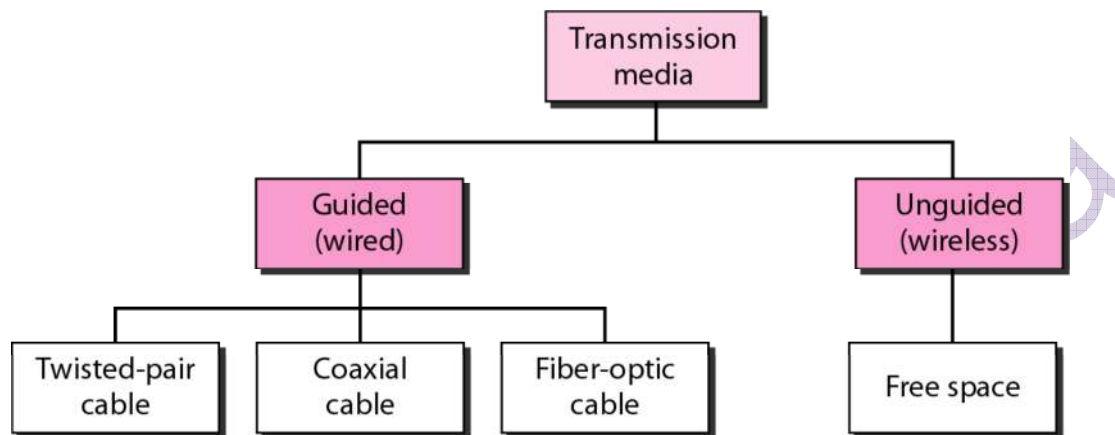
Static Routing — this type of routing requires manual feeding of data into the tables. Static routing is more errors during data entry and also takes longer time. This type of routing is suitable for small networks which have one or maximum two routers to avoid the errors.

Dynamic Routing — in this type of routing routers are enabled to transfer information to other routers so that all the routers get their own routing tables updated. Dynamic routing is preferable in larger network environments where more than two routers are in work.

Data Transmission Media

Transmission media is a pathway that carries the information from sender to receiver. We use different types of cables or waves to transmit data. Data is transmitted normally through electrical or electromagnetic signals. An electrical signal is in the form of current. An electromagnetic signal is series of electromagnetic energy pulses at various frequencies. These signals can be transmitted through copper wires, optical fibers, atmosphere, water and vacuum

Different Medias have different properties like bandwidth, delay, cost and ease of installation and maintenance. Transmission media is also called Communication channel. Transmission media is broadly classified into two groups.



Guided media: Guided media, which are those that provide a medium from one device to another, include twisted-pair cable, coaxial cable, and fiber-optic cable.

Twisted-pair cable: A twisted pair cable consists of two conductors (normally copper), each with its own plastic insulation, twisted together. One of the wires is used to carry signals to the receiver, and the other is used only as a ground reference.

Twisted Pair is of two types:

I. Unshielded Twisted Pair (UTP):-

An UTP cable consists of copper wires where each wire is separately encapsulated in PVC (polyvinyl chloride) insulation. It consists of color-coded copper wires, but does not include any foil or braiding as insulator to protect against interference.

The Electronic Industries Association (EIA) has developed standard to grade UTP cable by Quality; Category 1 as the lowest quality and category 6 as the highest quality.

1. Category 1: The basic twisted-pair cabling used in telephone systems. This level of quality is fine for voice but inadequate for data transmission.
2. Category 2: This category is suitable for voice and data transmission of up to 2Mbps.
3. Category 3: This category is suitable for data transmission of up to 10 Mbps. It is now the standard cable for most telephone systems. At least three twist per feet
4. Category 4: This category is suitable for data transmission of up to 20 Mbps.
5. Category 5: This category is suitable for data transmission of up to 100 Mbps.
6. Category 6: CAT- 6 is recently proposed cable type comprised of four pairs of wire capable of operating at transmission rates of up to 400Mbps.



Advantage:

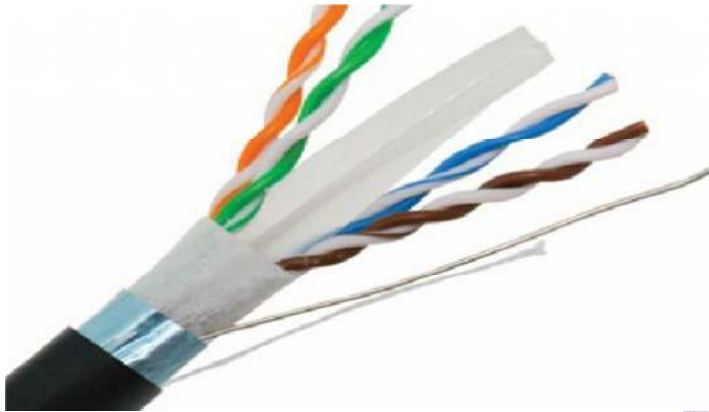
- It is smaller in size and light weight.
- It is less expensive compare to other networking media types.
- It is thin and flexible which further makes installation easier.
- UTP cables are used in most of the networking architecture.

Disadvantage:-

- It is more susceptible to interference compare to most of the other cable types.
- It can be used up to cable segment lengths of about 100 meters only.
- Lower capacity and performance in comparison to STP

ii .Shield twisted pair cable:-

STP cable has a metal foil or braided mesh covering that encases each pair of insulated conductors. Although metal casing improves the quality of cable by preventing the penetration of noise or crosstalk, it is bulkier and more expensive.



Advantages

- Easy installation.
- It has a higher capacity as compared to UTP cable.
- It is shielded that provides a higher data transmission rate.

Disadvantages

- Cables are bigger in diameter more expensive compared to UTP or coaxial cable.

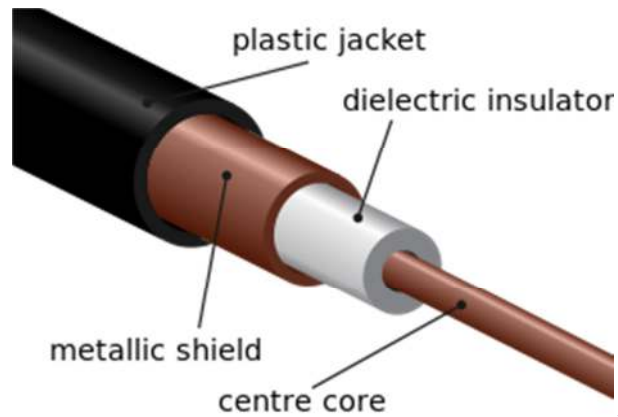
Applications

- Used as instrumentation cable in industrial applications.
- Used in harsh cold and hot environments.
- Research applications.

Coaxial Cable:- It is a wire made of copper which is in four layers. The middle part of the cable is made of thin wire. This wire is then covered with a layer of plastic.

The other two layers are that of the mesh. After that the topmost layer is made of rubber. Each end of the cable is covered with a thin layer of copper that we put in electrical equipment.

Coaxial (or “coax”) cable is a common type of cable used for transmitting data over long distances. It can carry either an analog or digital signal. While coax cables have many applications, they are most commonly used to transmit cable TV and Internet signals.



Advantage:

- Coaxial cable is very durable
- Best performance in short-distance transmission

Disadvantage:

- It is bulky.
- It is expensive to install for longer distances due to its thickness and stiffness.

Applications

- Coaxial cable was widely used in analog telephone networks, digital telephone networks
- Cable TV networks also use coaxial cables. Another common application of coaxial cable is in traditional Ethernet LANs.

Optical Fiber:-

A fiber-optic cable is made of glass or plastic and transmits signals in the form of light. Optical fiber works on the principle of total internal reflection. An optical fiber is a thin fiber of glass or plastic that can carry light from one end to the other. It works on the principle of total internal reflection. Optical cavities are designed to internally bend rays of light using TIR. The rays of light moving continuously strike the optical fiber wall and transmit the data from one end to the other.



Advantages:-

Large Bandwidth: Optical fiber is capable of covering long distances very quickly and in a very short time. These can have 10 to 100 times more bandwidth than power cables. Which means it is capable of bringing and carrying more data.

Electromagnetic interference: Optical fibers are made of insulators, so magnetic waves cannot affect them.

Small size and lightweight: Optical fibers are very small; their thickness is more than our hair. Due to this structure, their weight is also very less.

Security: Optical fiber provides complete safety to the data. It is not so cheap and easy to break its security.

Analog and digital signal transmission: Data is not transmitted in an analog way but in a digital way. Analog transmission can also be done in this. In analog transmission, the intensity varies continuously, so the use of digital mode is more appropriate.

Disadvantages:-

Expensive: – Optical fiber is more expensive than common data transfer cables.

Difficult setup: – The setup of optical fiber is very difficult.

Repair: - It is not easy to repair it. For this skill craftsmen are required.

Unguided Media:

In unguided Transmission Media data signals flows through the air. Wireless or Unguided Media or Unbound Transmission Media: Unbound transmission media are the ways of transmitting data without using any cables. These media are not bounded by physical geography. This type of transmission is called Wireless communication. Nowadays wireless communication is becoming popular. Wireless LANs are being installed in office and college campuses. This transmission uses Microwave, Radio wave, Infrared are some of popular unbound transmission media.

The following types of unguided media are:

Radio waves:-

Electromagnetic waves ranging in frequencies between 3 kHz and 1 GHz are normally called radio waves. Radio waves are Omni directional. When an antenna transmits radio waves, they are propagated in all directions. Radio waves use omnidirectional antennas that send out signals in all directions based on the wavelength, strength, and the purpose of transmission.

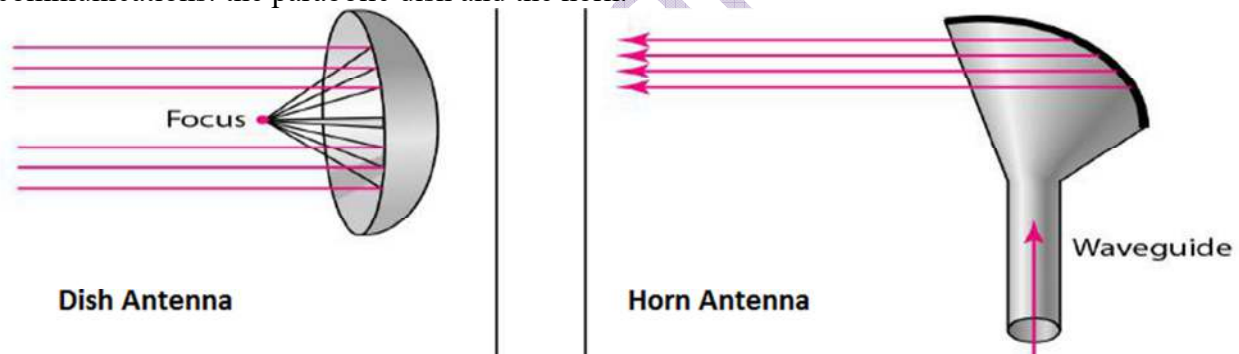


Applications

The Omni directional characteristics of radio waves make them useful for multicasting, in which there is one sender but many receivers. AM and FM radio, television, maritime radio, cordless phones, and paging are examples of multicasting.

Microwaves

Electromagnetic waves having frequencies between 1 and 300 GHz are called microwaves. Microwaves are unidirectional. The sending and receiving antennas need to be aligned. The unidirectional property has an obvious advantage. A pair of antennas can be aligned without interfering with another pair of aligned antennas. Microwaves need unidirectional antennas that send out signals in one direction. Two types of antennas are used for microwave communications: the parabolic dish and the horn.



Applications:

Microwaves are used for unicast communication such as cellular telephones, satellite networks, and wireless LANs.

Infrared:-

Infrared waves, with frequencies from 300 GHz to 400 THz (wavelengths from 1 mm to 770 nm), can be used for short-range communication. Infrared waves, having high frequencies, cannot penetrate walls. This advantageous characteristic prevents interference between one system and another; a short-range communication system in one room cannot be affected by another system in the next room.

Applications:

Infrared signals can be used for short-range communication in a closed area using line-of-sight propagation. Example Remote Control

Satellite:-

A satellite is an object that moves around a larger object. Earth is a satellite because it moves around the sun. The moon is a satellite because it moves around Earth. Earth and the moon are called "natural" satellites. There are thousands of man-made satellites.

How satellite works?

There are two solar panels attached on both sides. Transmitter and receiver are attached to send and receive signals. We can say that the satellite is made for communication only. It revolves around a satellite larger than itself, so the satellite is made in such a way that its speed does not allow the gravitational force of the Earth to dominate it. Due to which the satellite stays in the air.

Satellite is divided into three categories which are as follows.

Low Earth Orbit (LEO):

This is the nearest satellite from the earth. It can be up to an altitude of 160 km to 1600 km from the earth. This satellite is used for image or scanning. It completes many times revolution of earth in a day.

Medium earth orbit satellite:- It can be up to an altitude of 10 to 20 thousands km from the earth. This satellite is used for navigation. It completes one revolution of earth in 12 hours.

High earth orbit satellite:- It can be up to an altitude of 36000 thousands km from the earth. It completes one revolution with earth. This types of satellite is used for communication.

Advantages of Satellite:

The following are the advantages of satellite communication:

- It is used for mobile and wireless communication.
- It covers wide area of the earth hence entire country or region can be covered with just one satellite.
- It is easy to install and manage the ground station sites.

- It is used for voice, data, video and any other information transmission. Satellite system can be interfaced with internet infrastructure to obtain internet service. It is also used for GPS applications in various mobile devices for location determination.

Disadvantages of Satellite:

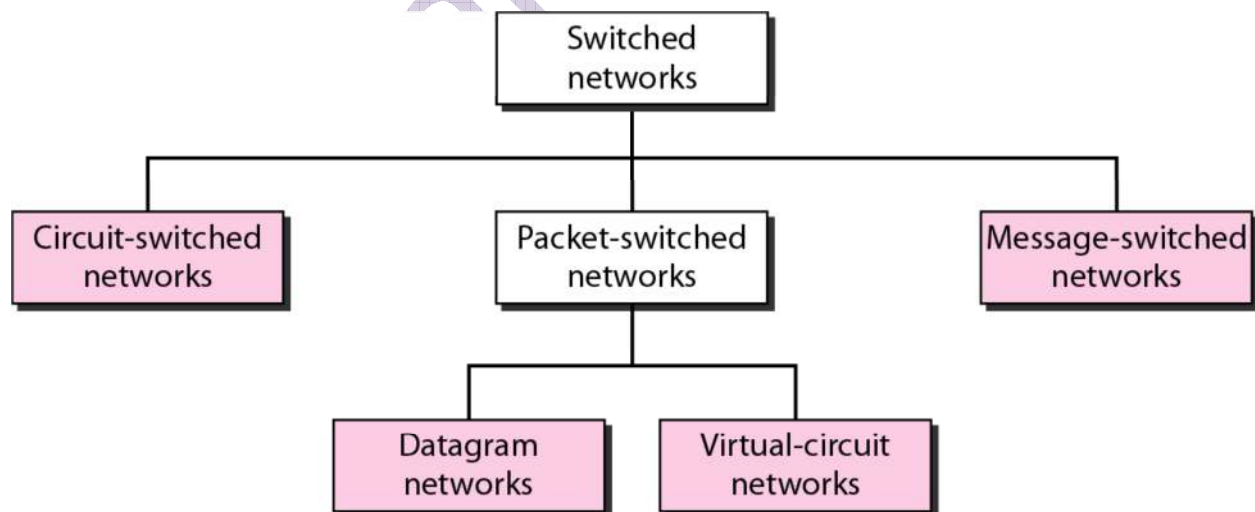
- Satellite manufacturing requires more time. Moreover satellite design and development requires higher cost.
- Satellite once launched, requires to be monitored and controlled on regular periods so that it remains in the orbit.
- Satellite has life which is about 12-15 years. Due to this fact, another launch has to be planned before it becomes un-operational.

Switch:

A switch is a mechanism that allows us to interconnect links to form a larger network. A switch is a multi-input, multi-output device that transfers packets from an input to one or more outputs. In the other words, “Switching is a process in which data or information is sent between different computer networks.” Switching technology is used in very large networks.

Types of Switching Techniques:

1. Circuit Switching
2. Packet Switching
3. Message Switching



1. Circuit Switching:-

When communication takes place between any two nodes through a given communication path then it is called circuit communication. In this, a pre-specified route is required for the transmission of data through which the given data travels, in which the travel of any other type of data is not permitted.

To transfer data in circuit switching, a circuit has to be established through which the data is transferred. Voice call is the best example of this in any telephone communication. In circuit switching, the circuit can be either permanent or temporary. Circuit switching passes through three phases:-

Circuit establishment

Data transfer

Circuit Disconnect

Advantages:

- A fixed path is made between sender and receiver.
- Data travels continuously after the connection is established.
- Suitable for telephone calls and real-time communication.
- Information follows the same path, so it does not get mixed up.

Disadvantages:

- Before talking, the network must create a path. Example: Like making a phone call. First, the call connects, and then we can talk.
- The path stays reserved for us even when we are silent. Example: During a phone call, if neither person speaks for 10 seconds, the line is still occupied.
- Each user gets a dedicated path.
- More cables and network resources are needed, so the cost is higher.

2. Packet switching:

In packet switching, the message is divided into smaller parts, these small parts of the message are called packets and each packet has its own source and destination address. And each packet is further transmitted in the network based on these addresses.

When all the packets reach the destination, then all of them are again converted into the original (original) message.

In packet switching, the network accepts the packets on the basis of FCFS (first come first serve) i.e. the packet which reaches first is served first. Internet is the best example of packet switching.

There are two types of packet switching:-

1. datagram packet switching:
2. virtual circuit packet switching

Datagram packet switching:

In datagram packet switching, each packet is transmitted independently in the network i.e. one packet does not have any relation with another packet. Being independent, the packet is called a datagram. These packets have a destination address from which they are transmitted in the network. In datagram packet switching, packets are independent because they are transmitted through different routes, due to which the packets reach the destination in a chaotic and bad manner. Datagram packet switching is also called connectionless packet switching.

Advantages:

- ✓ **No waiting to start**
 - We can send data immediately (no connection setup).
- ✓ **Fast use of network**
 - Network is used only when data is being sent.
- ✓ **If one path is blocked, another works**
 - Data can go through different roads.
- ✓ **Good for Internet**
 - Used in browsing, WhatsApp, YouTube, etc.

Disadvantages:

- ✓ **Data may come in wrong order**
 - First part may arrive later.
- ✓ **Some data may be lost**
 - All pieces may not reach.
- ✓ **Speed is not same always**
 - Sometimes fast, sometimes slow.
- ✓ **Needs fixing at end**
 - Receiver must arrange all parts correctly.

Virtual circuit packet switching:

In this type of packet switching, a route is selected between the sender and the receiver and all packets are transmitted through this route. Being transmitted through a single route, all the packets reach the destination systematically and correctly. In this, each packet is given its own number which is called virtual circuit number. Virtual circuit switching is done in the data link layer. Virtual circuit switching is also called connection oriented packet switching. Example of Virtual circuit packet switching is X.25, Frame Relay

Advantages

- ✓ **Data comes in order**
 - All packets follow the same path, so no mixing.
- ✓ **Less delay after setup**
 - After connection is made, data flows smoothly.
- ✓ **Efficient than circuit switching**
 - Network is shared with others.
- ✓ **Reliable communication**
 - Fewer errors compared to datagram switching

Disadvantages

- ✓ **Setup time needed**
 - First, a path must be created before sending data.
- ✓ **If path fails, connection breaks**
 - Whole communication stops if route has a problem.
- ✓ **Not fully flexible**
 - All packets must follow the same path.
- ✓ **Extra work for network**
 - In virtual circuit packet switching, the network (routers/switches) must remember the connection path for our data.

3. Message switching:

In message switching, there is no need to establish any special route between the sender and the receiver. In message switching, when a message is sent, it is accompanied by its destination address. In this, the message is transmitted from one node to another node. When the node receives the complete message, it stores it and when the second node is ready to receive the message, it forwards the message to it. E-mail is an example of a message switching system.

Advantages:

- ✓ **No dedicated connection required**
 - Sender and receiver do not need a fixed path.
- ✓ **Efficient use of network**
 - No line is reserved, so network resources are shared.
- ✓ **Message is stored safely**
 - If network is busy, message is stored and sent later.
- ✓ **Flexible routing**
 - Message can travel through different routes.

Disadvantages:

- ✓ **High delay**
 - Each node stores the full message before forwarding, so it is slow.
- ✓ **Requires large storage**
 - Routers need memory to store complete messages.
- ✓ **Not suitable for real-time communication**
 - Not good for voice calls or video.
- ✓ **Blocking problem**
 - If message is large, it can block the network for some time.

ISDN: interface and standard

Introduction to ISDN

ISDN (Integrated Services Digital Network) is a set of communication standards that allows voice, data, text, images, and video to be transmitted through a digital telephone network.

Before ISDN, telephone networks were mainly used for voice communication. ISDN introduced a digital network that could carry different types of information simultaneously.

Features of ISDN

- Fully digital communication system.
- Supports voice, data, fax, and video.
- Faster than traditional analog telephone systems.
- Better signal quality and less noise.
- Multiple services can use the same telephone line.

The ISDN has several kinds of access interfaces such as –

- Basic Rate Interface (BRI)
- Primary Rate Interface (PRI)
- Narrowband ISDN
- Broadband ISDN

Basic Rate Interface (BRI)

BRI (Basic Rate Interface) is an ISDN connection for **small users**.

- It is used in **homes and small offices**
- It works using normal telephone lines

BRI gives you 3 lines (channels):

2 Data/Voice Lines (B Channels)

- Used for calling or internet
- Speed: 64 kbps each

□ So:

- B1 = 64 kbps
- B2 = 64 kbps

1 Control Line (D Channel)

- Used for control work
- Example: dialing, connecting call, disconnecting call

Speed: 16 kbps

$$64 + 64 + 16 = 144 \text{ kbps}$$

Primary Rate Interface (PRI)

Primary Rate Interface (PRI) is an ISDN connection used for **large organizations** that need high-speed communication and many simultaneous calls.

- It is used in **banks, companies, offices, and call centers.**

Channel Structure of PRI

PRI also has B and D channels, but in a larger number.

B Channels (Bearer Channels)

- Used for **voice or data**
- Each channel = **64 kbps**
- Number of B channels:
 - **23B (T1 system – USA/Japan)**
 - **30B (E1 system – Europe/Asia)**

D Channel (Control Channel)

- Used for **signaling and control**
- Speed: **64 kbps**
- Only **1 D channel**

Total Speed

✓ T1 System:

$$23 \times 64 + 64 = 1.544 \text{ Mbps}$$

✓ E1 System:

$$30 \times 64 + 64 = 2.048 \text{ Mbps}$$

Narrowband ISDN

Narrowband ISDN (N-ISDN) is the traditional form of ISDN that uses **low-speed digital channels** to carry voice, data, and fax over the telephone network.

- It works using **64 kbps channels (B channels).**

Narrowband ISDN means:

- Low speed ISDN system
- Uses small bandwidth
- Based on **BRI and PRI**

Channel Structure

N-ISDN uses two main types of channels:

B Channel (Bearer Channel)

- Carries **voice and data**
- Speed: **64 kbps**

D Channel (Delta Channel)

- Carries **control and signaling**
- Speed:
 - 16 kbps (BRI)
 - 64 kbps (PRI)

Uses

- Home telephone services
- Small office communication
- Internet access (early days)
- Fax and voice calls

Broadband ISDN

Broadband ISDN (B-ISDN) is an advanced version of ISDN that provides very high-speed communication for transmitting voice, data, video, and multimedia services.

- It is designed for fast and large data transmission compared to Narrowband ISDN.
- B-ISDN = Very high speed (Mbps to Gbps range)

Technology Used in B-ISDN

ATM (Asynchronous Transfer Mode)

- Core technology of B-ISDN
- Transmits data in fixed-size cells (53 bytes)
- Supports fast switching and multiplexing
- Efficient for voice + video + data together

Features of Broadband ISDN

- Very high data transmission speed (Mbps to Gbps)
- Supports **multimedia services**
- Integrates voice, video, and data in one network

- Uses **ATM switching technology**
- Provides **real-time communication**
- Suitable for high bandwidth applications

What is an Interface?

An **interface** is a connection point between two devices or between a device and the network. It defines how information is exchanged between them.

The four main ISDN interfaces are:

1. R Interface
2. S Interface
3. T Interface
4. U Interface

1. R Interface

- Connects **non-ISDN device (TE2)** to **Terminal Adapter (TA)**
- Used for old devices

Example: normal computer, old phone

2. S Interface

- Connects **ISDN device (TE1)** to **NT2**
- Used inside homes/offices

Example: ISDN phone, ISDN fax

3. T Interface

- Connects **NT2 to NT1**
- Works inside network equipment

4. U Interface

- Connects **NT1 to ISDN service provider**
- This is the main telephone line

Assignment:

1. What is Network Media? Explain guided and unguided media in detail with examples.
2. Explain types of transmission media with neat diagrams.
3. Write short note on fiber optic cable.
4. Explain twisted pair cable and its types.
5. Explain ISDN and its features.

6. Describe ISDN interfaces (R, S, T, U).
7. Differentiate between BRI and PRI.
8. Explain Narrowband ISDN and Broadband ISDN in detail.

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